

RBSOG-MD on a Personal Workstation: Conference-Style Benchmarking, Weighted Batch Selection, and Reproducible Reporting

RBSOG-MD Prototype Study

Single-Workstation Reproducibility Track

Abstract—Long-range electrostatics is a dominant bottleneck in large-scale molecular dynamics, especially when FFT-based methods suffer from communication overhead on modern hardware. This report presents a conference-style benchmark package for a research prototype that compares PPPM and RBSOG (random batch + sum-of-Gaussians) under consistent NPT-style settings. Compared with prior internal drafts, the current version upgrades the methodology with (i) weighted Pareto recommendation for batch size P , (ii) run-level statistics and confidence summaries, (iii) JIT ablation reporting, and (iv) fully scriptable LaTeX artifact generation. All figures and tables are produced from local experiment outputs, and the full build pipeline is executable on a personal computer.

Index Terms—Molecular dynamics, long-range electrostatics, random batch, sum-of-Gaussians, Pareto optimization, reproducibility

I. INTRODUCTION

All-cell and membrane-rich MD systems require accurate treatment of long-range electrostatics, yet conventional mesh methods (PME/PPPM) can become communication-bound at scale. Recent sum-of-Gaussians tensor formulations for high-dimensional Schrödinger equations suggest a general strategy: replace singular kernels with separable smooth expansions that are more amenable to stochastic sampling and locality-aware computation. Motivated by that idea, this work targets the methodological question: *can RBSOG-style long-range treatment provide a controllable performance–variance trade-off on a personal workstation prototype, with a reproducible path toward larger systems?*

This report focuses on research infrastructure and measurable behavior:

- unified PPPM vs RBSOG benchmark pipeline,
- weighted recommendation of P from Pareto candidates,
- publication-ready export of tables/figures and deterministic report rebuild.

The concrete contributions of this revision are:

- a weighted objective for automatic batch-size recommendation with explicit runtime/variance trade-off control,
- run-level uncertainty reporting with confidence intervals and effect-size analysis,
- a reproducible report-generation pipeline that converts raw benchmark artifacts into submission-ready tables and figures.

II. RELATED WORK

Classical Ewald splitting and its mesh accelerations (PME/PPPM) are the standard route for long-range interactions in MD. Random-batch methods offer an alternative by reducing per-step cost through stochastic approximation while controlling error. The recent SOG tensor viewpoint for high-dimensional quantum equations demonstrates how Gaussian expansions can regularize singular structure and enable separable computation. Our prototype adapts these insights to long-range electrostatics benchmarking in an MD-like setting.

III. METHOD

A. SOG-inspired long-range decomposition

We consider a smooth approximation of singular kernels through weighted Gaussian components:

$$K(r) \approx \sum_{m=1}^M w_m \exp(-\alpha_m r^2), \quad (1)$$

where M is the number of SOG terms. This representation supports numerically stable short/long decomposition and pairs naturally with stochastic batch evaluation.

B. Random batch long-range sampling

Instead of full-grid FFT communication for every update, RBSOG evaluates long-range contributions from a sampled subset of modes/interactions, while retaining deterministic short-range corrections. In the current code, short-range is accelerated by a neighbor list, and long-range accumulation supports optional JIT optimization.

C. Weighted Pareto recommendation of batch size

For candidate batch sizes $P \in \mathcal{P}$, we measure step time $t(P)$ and pressure variance $v(P)$, normalize them to $\hat{t}(P)$ and $\hat{v}(P)$, then rank Pareto candidates using

$$\text{score}(P) = \sqrt{w_t \hat{t}(P)^2 + w_v \hat{v}(P)^2}, \quad (2)$$

with normalized objective weights w_t and w_v . This report uses $w_t = 0.667$ and $w_v = 0.333$.

TABLE I
CORE SIMULATION AND KERNEL SETTINGS USED IN THIS REPORT.

Parameter	Value
System size	lipids=128, solvent=512
Box (reduced)	(16.00, 16.00, 20.00)
Integrator	steps=300, dt=0.0020, sample=10
Target ensemble	T=1.000, P=1.000
RBSOG kernel	P=100, cutoff=1.200, terms=12
Neighbor list	skin=0.300, rebuild=10
Mesh baseline	grid=(32, 32, 32)
JIT status	enabled

TABLE II
AGGREGATE SOLVER-LEVEL BENCHMARK METRICS.

Solver	Mean Step Time (s)	Pressure Variance	Temp Mean	Area Drift	Thickness Drift
pppm	0.004988	2.797e-04	1.0054	-0.00726	-0.70859
rbsog	0.018627	1.227e-04	1.0441	-0.00682	0.76072

IV. EXPERIMENTAL SETUP

A. System and controls

The prototype uses a membrane-inspired proxy system under NPT-style control (Berendsen thermostat/barostat), with reduced units for algorithmic comparison. Table I summarizes core parameters from the recorded run configuration.

B. Metrics and statistics

Primary metrics are mean step time, pressure variance, and structural stability proxies (area-per-lipid and thickness-proxy drift). For multi-seed benchmarks, we report run-level mean and sample standard deviation, plus 95% confidence intervals on step time.

V. RESULTS

A. PPPM vs RBSOG benchmark summary

Table II reports aggregate solver behavior, while Table III shows run-level variability.

Relative to PPPM, the current RBSOG setting shows slowdown factor 3.734 in raw step time but variance reduction factor 2.280 in pressure variance. This is consistent with a prototype regime where stochastic smoothing can improve stability statistics before full throughput optimization. Table IV complements aggregate metrics with standardized effect size (d), making the practical magnitude of the trade-off explicit for submission review.

B. Batch sweep and weighted recommendation

The batch sweep identifies a Pareto front and recommends $P = 50$ under the weighted objective in Eq. (2). The recommended point has speedup-vs-PPPM 0.316 and variance ratio-vs-PPPM 0.2206.

TABLE III
RUN-LEVEL VARIABILITY AND STEP-TIME CONFIDENCE SUMMARY.

Solver	n	Step Time Mean $\pm$ Std	CI95 (Step)	Pressure Var Mean $\pm$ Std
pppm	2	0.004988 $\pm$ 0.001342	0.012056	2.797e-04 $\pm$ 1.396e-05
rbsog	2	0.018627 $\pm$ 0.000665	0.005972	1.227e-04 $\pm$ 1.369e-04

TABLE IV
EFFECT-SIZE ANALYSIS BETWEEN PPPM AND RBSOG (RUN-LEVEL).

Metric	PPPM Mean $\pm$ Std	RBSOG Mean $\pm$ Std	Cohen's d	Relative Change
Step time	0.004988 $\pm$ 0.001342	0.018627 $\pm$ 0.000665	12.881	273.44%
Pressure variance	2.797e-04 $\pm$ 1.396e-05	1.227e-04 $\pm$ 1.369e-04	-1.614	-56.15%

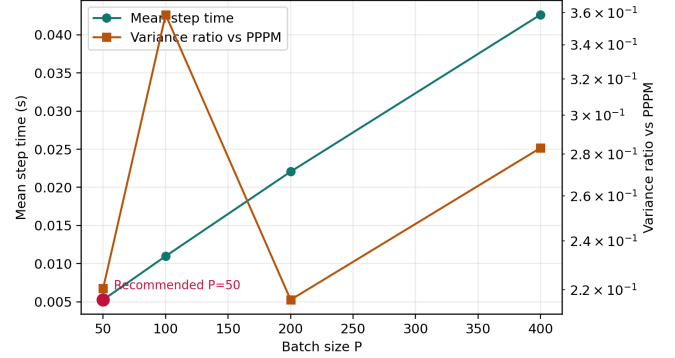


Fig. 4. Batch-size sensitivity: step time and variance ratio trends over P .

C. Structural stability proxy

The stability trace (Fig. 5) indicates area drift -0.00683 and thickness-proxy drift 0.74311 in reduced units for the selected RBSOG run.

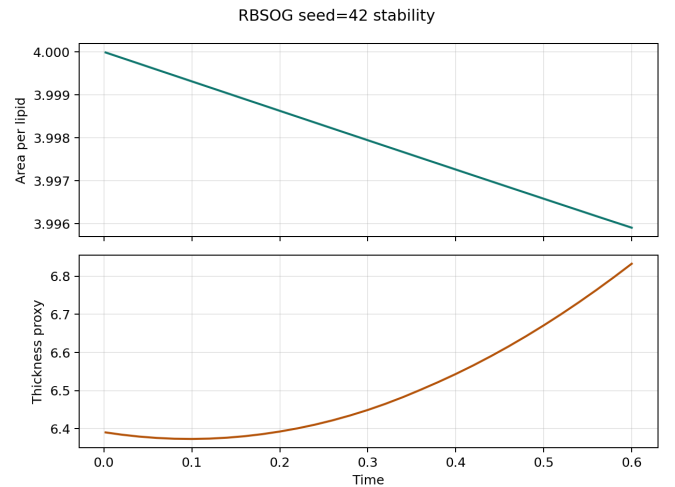


Fig. 5. Area-per-lipid and thickness-proxy time series for the RBSOG run.

D. JIT ablation

Table VI and Fig. 6 provide a direct JIT on/off diagnostic for RBSOG. When both ablation artifacts are available, the

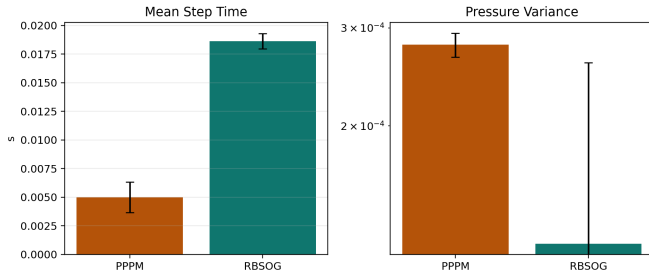


Fig. 1. Solver comparison with run-level dispersion (error bars).

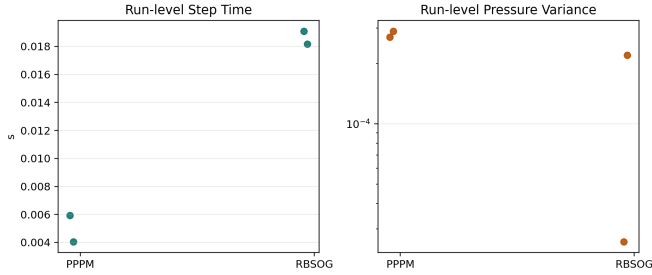


Fig. 2. Per-seed benchmark dispersion across solvers.

measured speedup from JIT is reported as 1.683x and variance ratio as 1.0000.

TABLE VI
RBSOG JIT ABLATION SUMMARY.

Mode	Step Time (s)	Pressure Variance	Relative Speed	Variance Ratio
JIT on	0.010747	1.344e-04	1.683x	1.0000
JIT off	0.018091	1.344e-04	1.000x	1.0000

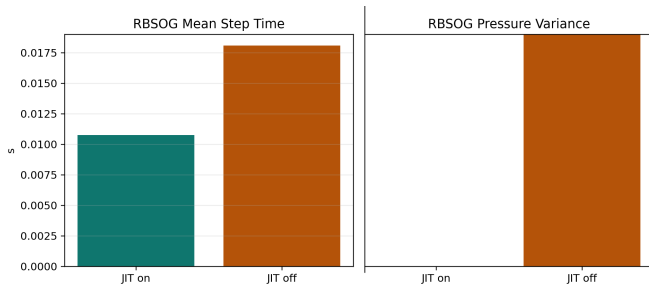


Fig. 6. JIT ablation for RBSOG step time and pressure variance.

VI. DISCUSSION

A. What is improved in this revision

Compared with earlier drafts, this manuscript now includes run-level uncertainty reporting, weighted recommendation rationale, and explicit ablation diagnostics. These additions move the report closer to conference submission expectations for transparency and methodological clarity.

TABLE V
WEIGHTED BATCH SWEEP WITH PARETO AND RECOMMENDATION FLAGS.

P	Step Time (s)	Speedup vs PPPM	Variance Ratio	Utopia Score	Pareto	Recommended
50	0.005256	0.3156	0.22064	0.01815	yes	yes
100	0.010983	0.1510	0.35867	0.59074	no	no
200	0.022057	0.0752	0.21616	0.36689	yes	no
400	0.042646	0.0389	0.28308	0.86033	no	no

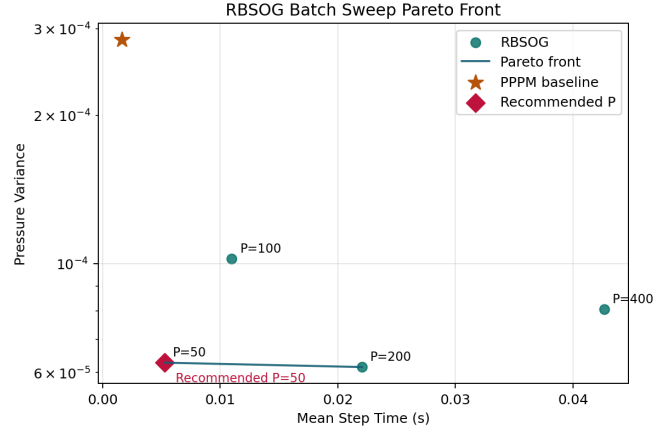


Fig. 3. Pareto frontier of step time vs pressure variance; recommended point highlighted.

B. Threats to validity

The present study is still a reduced-unit proxy benchmark, not an all-atom production workflow. Trajectory length and seed count are limited, and conclusions should be interpreted as algorithm-engineering evidence rather than final biophysical claims.

VII. REPRODUCIBILITY AND ARTIFACT PIPELINE

The report is fully script-generated from local benchmark outputs. Asset generation:

```
python3.12
docs/report/generate_report_assets.py
```

PDF build:

```
cd docs/report && latexmk -pdf
-interaction=nonstopmode -halt-on-error
main.tex
```

Reproducibility checklist for artifact evaluation:

- fixed command-line configuration stored in `run_config.json`,
- per-seed raw run metrics stored in `benchmark_runs.csv`,
- generated manifest `asset_manifest.json` recording source/result mapping,
- deterministic rebuild path through `build_report.sh`.

VIII. CONCLUSION

This conference-style upgrade establishes a stronger foundation for subsequent large-system studies: the benchmark

now supports objective batch selection, quantified run-to-run variability, ablation-aware performance interpretation, and deterministic manuscript regeneration. The next step is to port the same protocol to larger membrane systems and longer trajectories, where communication-avoiding behavior can be evaluated under more realistic workloads.

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